MM-RegCheck: Training-Free Causal-Aware Multimodal Reasoning for Regulatory Evasion Detection

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Abstract

Automated oversight of medical advertising requires authorization-based compliance checking rather than conventional truthfulness verification. This is especially challenging under cross-modal regulatory evasion, where benign text conceals images of unauthorized procedures. Existing vision-language models are brittle here, often over-relying on the safe modality and exploiting spurious correlations. We present MM-RegCheck, a training-free multimodal framework that extracts and grounds clinical procedures from image-text pairs using retrieved statutory evidence. To enforce compliance, we introduce TF-BEAR, a Bayesian evidence aggregation method. Using evidence-conditioned log-odds attribution, TF-BEAR isolates the regulatory evidence's contribution, enforcing strict liability when hidden visual evidence contradicts the text. We also introduce MedReg-MM, an expert-validated benchmark featuring adversarial evasion subsets. Using a 9B backbone, MM-RegCheck achieves 0.8985 Macro-F1 (Beauty) and 0.6556 (Healthcare), yielding state-of-the-art explanation quality (3.7 FIDES), and demonstrates robust cross-modality within-family generalization. These results highlight inference-time evidence adjudication as a practical foundation for high-stakes compliance monitoring.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Computer vision**.

Keywords

Causal Reasoning, Multimodal Regulatory Compliance Checking

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1 Introduction

The rapid proliferation of digital advertising on social media platforms has revolutionized digital marketing, yet it has catalyzed a public health crisis through the unchecked spread of non-compliant medical advertisements [5]. Unlike general misinformation, which often centers on the factual veracity of political or social claims [2], medical advertisements are strictly governed by complex, high-stakes legal frameworks. Consequently, automated oversight in this

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domain must fundamentally shift from traditional veracity checking to authorization-based regulatory compliance checking—verifying if a specific medical act is legally permitted to be advertised by a specific entity [12].

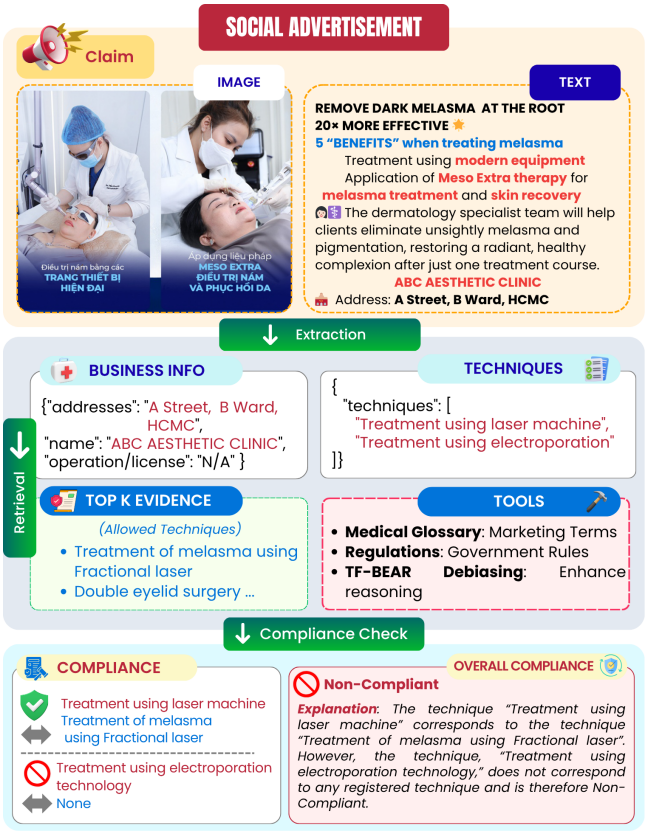


Figure 1: An illustrative example of the MM-RegCheck pipeline in action. *Note: Text is translated to English for demonstration; the original MedReg-MM dataset and regulatory corpus operate in Vietnamese.*

Modern advertising often employs ***Cross-Modal Evasion***: using legally "safe" text to mask unauthorized visual evidence (see Fig 1). Current Vision-Language Models (VLMs) and multimodal fusion frameworks are highly vulnerable to this [28]. Our preliminary evaluations reveal that standard backbones like Ministral-14B catastrophically collapse from a 0.7566 Macro-F1 under standard conditions to 0.4743 under evasion. They suffer from modality bias, erroneously anchoring predictions on the safe text, and exploit multimodal spurious correlations (e.g., predicting compliance solely based on a doctor's coat) [6, 31].

Existing paradigms exhibit critical flaws. Fine-tuning models on domain-specific datasets [34] lacks zero-day adaptability when regulations evolve [16]. Conversely, training-centric causal inference methods [20, 23] often fail to generalize to out-of-distribution evasion tactics in specialized legal contexts.

To overcome these limitations, we propose **MM-RegCheck**, a training-free, causal-aware agentic framework operating entirely at inference time. First, the ***Visual-Regulatory Projector*** bridges the semantic gap between visual metaphors and legal terminology by dynamically retrieving statutory definitions via Retrieval-Augmented Generation (RAG). Second, the ***Evidence Adjudicator*** employs the Training-Free Bayesian Evidence Aggregation for Reliable LLM Reasoning (TF-BEAR). Instead of simple feature fusion or token-level contrastive decoding, TF-BEAR performs macro-level, evidence-conditioned log-odds attribution (Δ) across reasoning trajectories. By isolating the true informational gain of visual evidence against retrieved laws, it establishes a strict dual-threshold gate to prune hallucinated branches, rigorously enforcing a "strict liability constraint" where visual violations unequivocally override deceptive text.

Our primary contributions are threefold: (1) Formulating Multimodal Regulatory Evasion and introducing **MedReg-MM**, a benchmark with adversarial "Hard Evasion" subsets. (2) Proposing MM-RegCheck, uniquely integrating RAG with inference-time Bayesian evidence aggregation to mitigate modality bias without retraining. (3) Achieving SOTA performance (0.8985 Macro-F1) with a highly efficient 9B backbone, demonstrating profound adversarial robustness.

2 Literature Review

Early MFC systems addressed political misinformation and out-of-context image-text pairs [28, 36]. Recent architectures like MOCHEG [34] integrate evidence retrieval, while frameworks like DEFAME advance single-pass verification [29]. However, these assume modality symmetry, optimizing for veracity rather than compliance. Consequently, they remain vulnerable to Cross-Modal Evasion, where safe text suppresses visual violations. MM-RegCheck replaces symmetric fusion with a conflict-sensitive reasoning chain that explicitly prioritizes visual violations.

2.1 Causal Inference in Vision-Language Models

Vision-Language Models (VLMs) notoriously suffer from multimodal spurious correlations, such as relying on a "doctor's coat" to predict compliance rather than engaging in genuine semantic reasoning [6]. To mitigate this, causal inference [25] has been increasingly adopted to decouple causal features from confounding noise. In visual question answering, Niu et al. [23] introduced counterfactual reasoning to reduce language bias. More recently, in the fact-checking domain, the CCMC framework [20] formulated dataset biases as causal effects, employing causal intervention to construct counterfactual instances to train debiased sub-checkers.

While theoretically profound, current causal interventions in MFC share a critical bottleneck: they are strictly training-centric. Furthermore, inference-time contrastive methods, such as Contrastive Decoding (CD) or Pointwise Mutual Information (PMI) based penalizations, typically operate at the *micro token-decoding level* (e.g., contrasting $P(y|x,c) - P(y|x)$) to continuously penalize generic text generation.

MM-RegCheck bypasses these limitations by introducing a macro-level approach. By leveraging an inference-time Bayesian evidence aggregation framework (TF-BEAR), our approach explicitly computes the evidence-conditioned log-odds attribution (Δ) of visual evidence across entire reasoning trajectories. Unlike continuous token penalties in CD, TF-BEAR utilizes Δ as a strict dual-threshold

Table 1: Comparative analysis of MedReg-MM against prominent multimodal fact-checking benchmarks.

Benchmark Dataset	Domain	Verification Target	Explanation	Adversarial Subsets	Samples
Fakeddit [22]	Reddit/ Social Gossip	Fake News Classification	✗	✗	~1M
Factify [27]	General News	Veracity & Entailment	✗	✗	~50k
MOCHEG [34]	Politics/ News	Veracity	✓	✗	~15k
MMM-Fact [33]	Multiple domains	Veracity	✓	✗	~125k
MedReg-MM (Ours)	**Medical/ Aesthetics**	**Compliance (Authorization)**	✓	✓	**~6k**

gate to prune hallucinated logical branches entirely. This "plug-and-play" mechanism requires no gradient updates, offering a robust and highly scalable alternative to computationally expensive, training-heavy counterfactual pipelines.

2.2 Agentic & RAG-based LLMs/VLMs

Retrieval-Augmented Generation (RAG) has emerged as the de facto standard for mitigating hallucinations and injecting external, domain-specific knowledge into Large Language Models [18]. While highly effective in text-to-text question answering, the application of RAG in multimodal domains—particularly for strict legal and regulatory mapping—remains underexplored. VLMs, including cutting-edge Mixture-of-Experts architectures, exhibit impressive zero-shot descriptive capabilities but fundamentally lack the granular legal vocabulary required to map high-level visual metaphors to specific regulatory statutes [26].

Recent works have explored LLM-based agents capable of utilizing external tools and knowledge bases to perform multi-step reasoning [32]. Building upon this paradigm, MM-RegCheck introduces the Visual-Regulatory Projector. Instead of relying on generic image captioning, our framework dynamically retrieves statutory definitions from a vectorized legal knowledge base. This retrieved context acts as an instructional anchor, guiding the frozen VLM to actively search the image for legally prohibited elements rather than merely describing its contents. This continuous, training-free alignment between visual evidence and legal knowledge empowers MM-RegCheck to instantly adapt to new regulatory constraints.

2.3 Multimodal Fact-Checking Benchmarks

The evolution of AFC is deeply intertwined with the development of large-scale benchmark datasets. Early datasets, such as MultiFC [3], laid the foundation for evidence-based verification but were limited to unimodal text. The subsequent shift toward multimedia resulted in prominent datasets like Fakeddit [22], Factify [27], and MOCHEG [34], which incorporate image-text pairs to assess out-of-context misinformation and multimodal fake news.

However, a critical review of these benchmarks reveals two significant gaps. First, they are overwhelmingly oriented toward general news, politics, and social media gossip, focusing exclusively on factual veracity. They lack the domain-specific rigor required for high-stakes regulatory environments, where the objective is authorization-based compliance grounded in explicit legal statutes. Second, existing datasets predominantly consist of straightforward, naturally occurring misinformation. They critically lack meticulously constructed adversarial subsets designed to stress-test a model's causal reasoning against intentional "Cross-Modal Evasion" tactics. Table 1 illustrates these critical distinctions, highlighting the necessity of MedReg-MM as the first benchmark designed to evaluate multimodal regulatory compliance and reasoning robustness against deliberate semantic camouflage.

3 PROBLEM FORMULATION & DATASET

To systematically evaluate regulatory compliance in a multimodal context, we formalize the problem space and introduce **MedReg-MM**, the first large-scale, expert-validated benchmark specifically engineered to test the causal reasoning limits of Vision-Language Models (VLMs) against cross-modal evasion. Detailed operational documentation, including Geographic Scope Justification, Annotation Guidelines, and the comprehensive Dataset Datasheet, is deferred to Technical Appendices A through D.

3.1 Formal Definition

Unlike traditional veracity-based fact-checking [30], regulatory compliance verification focuses on assessing whether an advertisement adheres to legally authorized scopes of practice under domain-specific regulations [29].

We consider a multimodal advertisement represented as $\boldsymbol{A} = (\boldsymbol{T}, \boldsymbol{I})$, where $\boldsymbol{T}$ denotes textual claims and $\boldsymbol{I}$ denotes visual evidence (image). The advertisement promotes a set of techniques $\boldsymbol{E_T} = \{\boldsymbol{t_1}, \boldsymbol{t_2}, \ldots, \boldsymbol{t_n}\}$ by an entity $\boldsymbol{e}$. The regulatory context varies by domain: in the ***Beauty*** domain, compliance is determined based on the entity's authorized scope of practice $\boldsymbol{S_e}$ and a global regulatory framework $\boldsymbol{R}$ specifying permitted and prohibited procedures; in the ***Healthcare*** domain, compliance is governed by the set of registered and licensed techniques $\boldsymbol{R_e}$ approved for the entity. We unify these as a context variable $\boldsymbol{C}$, where $\boldsymbol{C} = (\boldsymbol{S_e}, \boldsymbol{R})$ for Beauty and $\boldsymbol{C} = \boldsymbol{R_e}$ for Healthcare.

The task is to learn a mapping function $\boldsymbol{f} : (\boldsymbol{T}, \boldsymbol{I}, \boldsymbol{C}) \rightarrow (\boldsymbol{y}, \boldsymbol{X})$, where $\boldsymbol{y} \in \{\textbf{Compliant}, \textbf{Partly-Compliant}, \textbf{Non-Compliant}\}$ denotes the compliance verdict and $\boldsymbol{X}$ is a natural language explanation. Compliance is first assessed at the level of individual techniques $\boldsymbol{t_i} \in \boldsymbol{E_T}$, then aggregated into an overall verdict via a strict hierarchy: the advertisement is classified as ***Non-Compliant*** if any technique is non-compliant; otherwise, it is ***Partly-Compliant*** if any technique is partially compliant; and ***Compliant*** only if all techniques satisfy regulatory requirements. This formulation captures both fine-grained reasoning at the technique level and holistic decision-making at the advertisement level.

The Strict Liability Constraint: We further impose a strict liability constraint reflecting real-world regulatory practice, where visual evidence takes precedence over textual claims. Specifically, if the image $\boldsymbol{I}$ depicts a prohibited or unauthorized procedure

p, the advertisement must be classified as ***Non-Compliant***, regardless of whether the textual content is compliant or evasive. Formally, this cross-modal dominance can be expressed as $P(y = \textbf{Non-Compliant} \mid I_{\text{violating}}, T_{\text{safe}}, C) = 1$, where a violation occurs when $p \notin S_e$ or contradicts R in the Beauty domain, and when $p \notin R_e$ in the Healthcare domain. This constraint introduces an inherent asymmetry in multimodal reasoning, requiring the model to prioritize visual signals when inconsistencies arise. Current standard multimodal fusion architectures fail to satisfy this constraint, allowing T_{safe} to suppress the causal signal of $I_{\text{violating}}$ [28].

3.2 The MedReg-MM Benchmark

MedReg-MM comprises 5,706 real-world multimodal advertisements deterministically paired with an authoritative corpus of 9,978 registered legal facilities.

3.2.1 Base Dataset Construction and Validation. The base dataset targets highly regulated aesthetic domains. To ensure ground-truth fidelity at scale, we employed a hybrid Human-AI annotation protocol [9]. An instruction-tuned LLM provided preliminary claim extractions and compliance labels by cross-referencing advertisements with the official registry R.

To mitigate AI hallucination risks, these preliminary labels underwent rigorous human evaluation. A panel of domain experts (specializing in medical law) audited a 10% Gold Standard (GS) subset (600 samples). The inter-annotator agreement yielded a substantial Krippendorff's $\boldsymbol{\alpha = 0.6386}$ for multimodal judgments. Following structured deliberation on edge cases, experts reached an almost perfect consensus of **Cohen's $\boldsymbol{\kappa = 0.9200}$**, establishing an exceptionally reliable foundation for the benchmark.

3.2.2 Adversarial Subsets via Constructive Intervention. To stress-test VLMs against intentional deception, we utilized a *Constructive Intervention* methodology [17] to engineer two adversarial subsets from the non-compliant base data ($I_{\text{violating}}$).

The Evasion Subset (Textual Manipulation): This subset operationalizes the "Trojan Horse" tactic observed in real-world regulatory evasion, where advertisers obscure prohibited medical procedures behind cosmetically safe language. Using LLM-guided text style transfer, we systematically rewrote the violating post content for each of the 200 Non-Compliant samples, replacing intrusive medical terminology with legally ambiguous commercial euphemisms and leaving the visual evidence unchanged. Each resulting counterfactual pair (T_{safe}, $I_{\text{violating}}$) retains the ground-truth label of Non-Compliant, since the depicted procedure remains prohibited regardless of how the accompanying text is framed.

The Conflict Subset: This subset evaluates robustness against bad-faith legal disclaimers. We preserved the violating image but appended explicit negation clauses to it (e.g., "*Disclaimer: The image is for illustrative purposes only*"). This counterfactual pair (T_{safe}, $I_{\text{violating}} \oplus$ Disclaimer) acts as the ultimate test of the Strict Liability Constraint, requiring the model to recognize that visual evidence supersedes textual negation.

3.3 Analytical Properties and NLP Challenges

MedReg-MM is partitioned into validation (50%) and test (50%) splits, with adversarial hard subsets reserved exclusively for the test set to ensure zero-shot evaluation and prevent data leakage. This setup enforces robust generalization. Dataset analysis further reveals key challenges for NLP systems.

Table 2: Average semantic similarity.

Compliance	Avg. Sim.
Compliant	0.609
Partly-Compliant	0.575
Non-Compliant	0.443

Semantic Gap. A fundamental challenge of the benchmark lies in bridging the semantic gap between informal, often persuasive advertising language and the precise terminology defined in regulatory frameworks. To quantify this phenomenon, we compute the semantic similarity between advertised techniques and their corresponding permitted scopes. As summarized in Table 2, Non-Compliant instances exhibit a substantially lower average similarity score (0.443) compared to Compliant (0.609) and Partly-Compliant (0.575) cases. Furthermore, the majority of similarity scores are concentrated within a relatively narrow range (approximately 0.4–0.6), indicating that surface-level lexical overlap is insufficient for reliable classification. These findings demonstrate that the task requires models to perform deep semantic alignment and contextual reasoning, rather than relying on naive keyword matching strategies.

4 THE MM-REGCHECK FRAMEWORK

To address the profound challenges of cross-modal evasion and zero-day adaptability in regulatory compliance, we propose **MM-RegCheck**, a training-free, Bayesian framework.

Designed to operationalize the compliance mapping function $f : (T, I, C) \rightarrow (y, X)$ formulated in Section 3, the framework diverges from conventional "black-box" multimodal fusion paradigms that are highly susceptible to spurious correlations [28]. MM-RegCheck executes this deterministic evaluation through two conceptually distinct pillars: the *Visual-Regulatory Projector* for precise semantic alignment, and the *Evidence Adjudicator* for rigorous, hallucination-resistant logical inference. The overall architecture is illustrated in Figure 2, while detailed operational implementations, including prompt engineering strategies, are deferred to Appendix E.

4.1 Pillar 1: Visual-Regulatory Projector

The primary vulnerability of generic Vision-Language Models (VLMs) in specialized legal domains is the semantic gap; they tend to generate broad, descriptive captions rather than extracting explicit, legally actionable clinical procedures, thereby precipitating object hallucination [6]. The Visual-Regulatory Projector overcomes this limitation by anchoring the visual extraction process in a standardized medical taxonomy and dynamically linking it to statutory constraints via Retrieval-Augmented Generation (RAG) [18].

The projection initiates with concept-guided clinical extraction. Instead of unconstrained image captioning, the frozen VLM is prompted to act as a medical domain expert. Guided by a predefined medical glossary (see Appendix G), the model interrogates the multimodal input to aggressively filter out euphemistic marketing claims. It normalizes both visual and textual cues into a discrete set of canonical, physical clinical techniques E_T (detailed prompt constraints are provided in Appendix E).

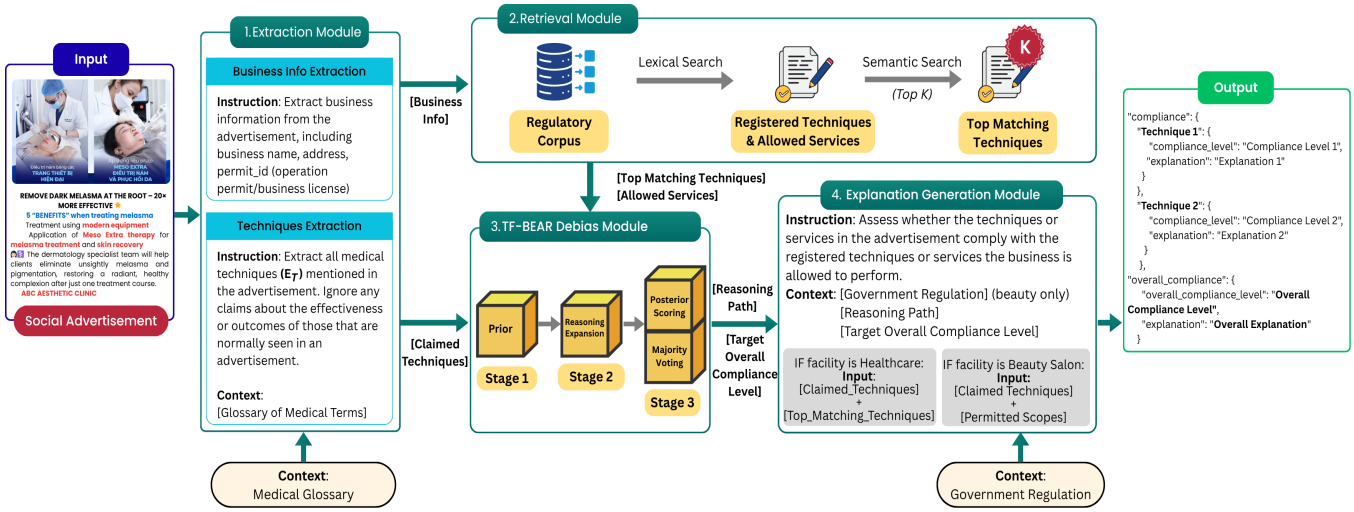


Figure 2: The overall architecture of the proposed framework. It operates through four sequential modules: (1) Extraction Module for identifying business information and claimed medical techniques from multimodal inputs, (2) Retrieval Module for fetching registered techniques via lexical and semantic search, (3) TF-BEAR Debias Module for robust prior-posterior reasoning, and (4) Explanation Generation Module for synthesizing the final verdict and comprehensive explanation.

Following extraction, the framework executes knowledge-anchored semantic retrieval. To bridge the remaining gap between the extracted colloquial claims $\boldsymbol{E_T}$ and formal legal language, we employ a dense vector retrieval mechanism against the domain-specific regulatory context $\boldsymbol{C}$. Each extracted technique $\boldsymbol{t_i}$ is embedded into a high-dimensional semantic space and compared against the embedded representations of all registered statutory provisions. As elaborated in Appendix F, this mechanism applies a Top-k filtering strategy to retrieve the most semantically relevant legal permissions, forming a highly focused evidentiary subset $\hat{\boldsymbol{C}}$. This pivotal step transforms the intrinsically open-ended reasoning task into a closed-book, evidence-grounded evaluation.

4.2 Pillar 2: Evidence Adjudicator via TF-BEAR

Even with precisely retrieved evidence $\hat{\boldsymbol{C}}$, standard Large Language Model (LLM) reasoning pathways remain highly vulnerable to modality bias. A model might correctly retrieve a legal statute but still emit an erroneous verdict by over-trusting deceptive textual claims $\boldsymbol{T}$ or being confounded by spurious visual contexts within $\boldsymbol{I}$ (e.g., a practitioner's authoritative attire). To mathematically enforce the strict liability constraint, we propose **Training-Free Bayesian Evidence Aggregation for Reliable LLM Reasoning (TF-BEAR)**. This engine shifts the decision-making process from generating subjective text tokens to evaluating strictly calibrated, internal model probabilities via evidence-conditioned log-odds attribution.

The TF-BEAR algorithm operates through three interconnected phases. **Initially, through prior probability elicitation**, we extract the model's baseline belief regarding the compliance of the techniques $\boldsymbol{E_T}$ before exposing it to the regulatory evidence $\hat{\boldsymbol{C}}$. To prevent uncalibrated generative hallucination, we constrain the LLM API to output a single token corresponding to the classification labels and directly extract the unnormalized log probabilities (`logprobs`). Applying a softmax function yields a rigorously calibrated prior probability distribution, $P_{prior}(\boldsymbol{y} \mid \boldsymbol{E_T})$, which quantitatively captures the model's inherent susceptibility to the advertisement's euphemistic framing.

Subsequently, during evidence-conditioned posterior synthesis, the reasoning space is aggressively expanded utilizing various prompting methods [32, 35]. Each generated reasoning branch integrates the extracted techniques $\boldsymbol{E_T}$ with the authoritative legal evidence $\hat{\boldsymbol{C}}$, forcing the model to explicitly evaluate any contradictions between the visual act and the law. For each terminal branch, a second direct `logprobs` extraction is performed to compute the posterior probability distribution, $P_{posterior}(\boldsymbol{y} \mid \boldsymbol{E_T}, \hat{\boldsymbol{C}})$.

Ultimately, Bayesian log-odds filtering establishes the definitive verdict through a mathematical safeguard against hallucination [20]. We calculate the log-odds delta (Δ) for each reasoning branch, representing the true informational gain exclusively provided by the legal evidence. Operating in log-odds space, defined as $L(p) = \log(p/(1-p))$, the evidence-conditioned attribution is computed as:

$$\Delta = L(P_{posterior}(\boldsymbol{y})) - L(P_{prior}(\boldsymbol{y}))$$

To be incorporated into the final synthesis, a reasoning branch must satisfy a strict dual-threshold condition: high absolute posterior confidence ($P_{posterior}(\boldsymbol{y}) \geq \tau$) and substantial evidence-driven belief revision ($\Delta \geq \tau$). Branches failing this mathematical filter are aggressively pruned. The Adjudicator then derives the technique-level compliance via a majority vote among the surviving, causally sound branches (formalized in Appendix H), which are finally aggregated into the overall advertisement verdict $\boldsymbol{y}$.

Table 3: Comprehensive performance evaluation of MM-RegCheck against SOTA baselines. Results report metrics across standard (Open-Book), ungrounded (Close-Book), and adversarial (Hard Evasion) conditions in the Beauty and Healthcare domains. Bold indicates the best performance, while underline denotes the second-best.

Model	Open-Book			Close-Book			Hard Evasion		
	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall
Beauty									
Phi-4 (14B)	0.7459	0.7626	0.7544	0.3124	0.4572	0.2444	0.5388	0.8736	0.5625
Ministral (14B)	0.7566	0.8558	0.6931	**0.4430**	**0.8931**	0.4998	0.4743	0.4560	0.4940
Llama-4-Maverick (17B)	0.7832	0.7857	0.8549	0.2341	0.6354	**0.5055**	0.6082	0.8953	0.6000
Qwen3-VL (8B)	0.8685	0.8432	0.9170	0.4261	0.4470	0.4268	**0.6770**	**0.9560**	**0.6250**
Qwen3.5 9B (Ours)	**0.8985**	**0.8779**	**0.9283**	0.4197	0.4277	0.4146	0.6186	0.9341	0.5909
Healthcare									
Phi-4 (14B)	0.5579	0.5798	0.5849	0.2748	0.2431	0.3213	0.2313	0.4262	0.3838
Ministral (14B)	**0.6702**	0.6683	**0.6893**	0.2340	0.1803	0.3333	0.3353	0.5301	0.3515
Llama-4-Maverick (17B)	0.6485	0.6582	0.6782	0.2305	**0.3770**	0.3068	0.5246	0.6778	0.4932
Qwen3-VL (8B)	0.5654	0.6511	0.6418	**0.3303**	0.3089	**0.3724**	0.3707	0.5902	0.3598
Qwen3.5 9B (Ours)	0.6556	**0.6799**	0.6820	0.2826	0.2720	0.3379	**0.5742**	**0.7177**	**0.7134**

5 Experiments and Results

5.1 Experimental Setup

We evaluate all models on the **MedReg-MM** test split, encompassing the standard distribution and the adversarial "Hard Evasion" subset defined in Section 3. To measure classification performance for the discrete verdict $\boldsymbol{y}$, we report Macro-F1, Precision, and Recall. To evaluate the qualitative coherence of the generated explanation $\boldsymbol{X}$, we employ the multi-dimensional FIDES-Score (Fidelity, Justification, Clarity), a robust metric adapted from advancements in explainable fact-checking evaluation [7, 13, 29].

To ensure a comprehensive evaluation, our baselines include Llama-4-Maverick (17B) [11], Phi-4 (14B) [1], Ministral (14B) [15], and Qwen3-VL (8B) [4]. Crucially, for a fair comparison, all baselines are integrated as backbones within the MM-RegCheck framework, utilizing the identical TF-BEAR adjudicator and RAG context. For our proposed framework, we instantiate MM-RegCheck using Qwen3.5 (9B) as the core reasoning backbone, chosen for its optimal balance of deep reasoning capabilities and computational efficiency. Details are in Appendices I-J.

5.2 Main Results and Analysis

Table 3 summarizes the overall performance across the *Beauty* and *Healthcare* domains under Open-Book (evidence-grounded) and Close-Book conditions. In the Beauty sector, MM-RegCheck, powered by the Qwen3.5 (9B) backbone, establishes a definitive SOTA with an exceptional Macro-F1 of 0.8985 under Open-Book conditions. This significantly outperforms computationally heavier backbones like Llama-4 (17B) (0.7832) and Ministral (14B) (0.7566).

Crucially, the experiment reveals a significant domain disparity. Performance across all baseline models decays substantially when transitioning from the Beauty to the highly regulated Healthcare domain. This decay is directly attributable to the hyper-granularity of the medical regulatory context $\boldsymbol{C}$, where latent parametric knowledge becomes increasingly unreliable [26]. While Ministral achieves the highest Open-Book F1-score (0.6702) in Healthcare—likely leveraging its larger parameter count for parametric memorization—our efficient 9B framework remains highly competitive (0.6556). Furthermore, the stark contrast between the Open-Book and Close-Book configurations unequivocally proves that latent knowledge is entirely insufficient for strict authorization tasks. When deprived of the retrieved evidence $\hat{\boldsymbol{C}}$ in the Close-Book setting, the backbone model's performance in the Beauty domain plummets catastrophically to an F1-score of 0.4197, while Llama-4 falls to 0.2341, reinforcing the absolute necessity of our Visual-Regulatory Projector.

Robustness against Cross-Modal Evasion. The ultimate test of a compliance framework is its resilience against adversarial deception [10]. In the "Hard Evasion" subset, the textual modality $\boldsymbol{T}$ is intentionally manipulated to project a legally safe image, while the visual modality $\boldsymbol{I}$ depicts a clear violation. Without evidence adjudication,models heavily prioritize the safe text and violate the strict liability constraint. For instance, Ministral, despite its strong Open-Book performance, catastrophically crashes to F1-scores of 0.4743 (Beauty) and 0.3353 (Healthcare) under adversarial conditions.

Conversely, MM-RegCheck demonstrates profound causal robustness. It achieves a highly competitive F1-score of 0.6186 in the Beauty adversarial subset (second only to the vision-optimized Qwen3-VL), and establishes a definitive SOTA in the highly complex Healthcare adversarial subset with an F1-score of 0.5742, outperforming Llama-4 (17B) by a substantial margin. By extracting the prior probability $P_{prior}(\boldsymbol{y} \mid \boldsymbol{E}_T)$ before evidence injection, the Evidence Adjudicator mathematically isolates and mitigates the deceptive textual bias. The log-odds framework ensures that the final verdict is driven by the causal impact of the retrieved visual violations rather than the deceptive textual context, validating the theoretical advantages of causal reasoning in multimodal tasks [23].

5.3 Ablation Study

To mathematically quantify the contribution of our core pillars, we conducted a rigorous ablation study on our Qwen3.5 backbone, the results of which are detailed in Table 4. First, bypassing the evidence retrieval module entirely causes a catastrophic performance collapse, evidenced by a massive F1-score reduction (ΔF1 = 0.4259). This verifies that dynamic legal grounding ($\hat{C}$) serves as the absolute foundational requirement of the framework. Second, removing the structured regulatory guidelines from the context leads to a severe drop (ΔF1 = 0.1147), highlighting the critical necessity of explicitly defining domain boundaries for the reasoning agent.

Table 4: Ablation study on the core components of MM-RegCheck. The Delta F1-Score highlights the performance drop when each specific module is removed.

Configuration	F1-score		Δ F1
	Beauty	Healthcare	
Full Framework	**0.8985**	**0.6556**	**0.0000**
zero-shot	0.8650	0.5335	0.0778
w/o Expansion	0.8746	0.6434	0.0181
w/o Reg. Knowledge	0.7838	-	0.1147
w/o Retrieval	0.4197	0.2826	0.4259

* Reg. Knowledge is only used for the Beauty domain.

Furthermore, substituting our Bayesian log-odds filter (TF-BEAR) with standard zero-shot prompting results in a notable decline (ΔF1 = 0.0778). This confirms that TF-BEAR is explicitly responsible for eliminating multimodal hallucinations that standard generative pathways fail to catch [14]. Finally, removing the reasoning expansion step marginally decreases performance (ΔF1 = 0.0181), demonstrating that while the prior-posterior log-odds delta remains the primary driver of accuracy, expanding the reasoning space provides necessary stabilization for complex legal deductions.

5.4 Case study

To concretely illustrate the efficacy of inference-time evidence adjudication, we examine a representative instance from the MedReg-MM subset characterized by domain-specific jargon obfuscation. In this scenario, the visual modality depicts entirely benign before-and-after acne treatment outcomes, while the textual modality embeds restricted medical procedures (e.g., "TCA Peel") within routine cosmetic marketing vernacular.

Standard VLM systematically succumb to this form of evasion. Anchored by modality bias toward the safe visual cues and lacking granular legal knowledge, baselines erroneously yield a "Compliant" verdict with a high prior confidence (P_{prior} = 61.7%). MM-RegCheck mathematically deconstructs this oversight. The framework's extraction module normalizes the embedded jargon into canonical clinical interventions (i.e., chemical peeling) and aligns it with the prohibitive statutory constraints ($\hat{C}$) specific to basic skincare facilities. During expansion, the framework samples a diverse set of reasoning paths instead of only taking the greedy one, enabling it to select the most consistent answer across multiple reasoning

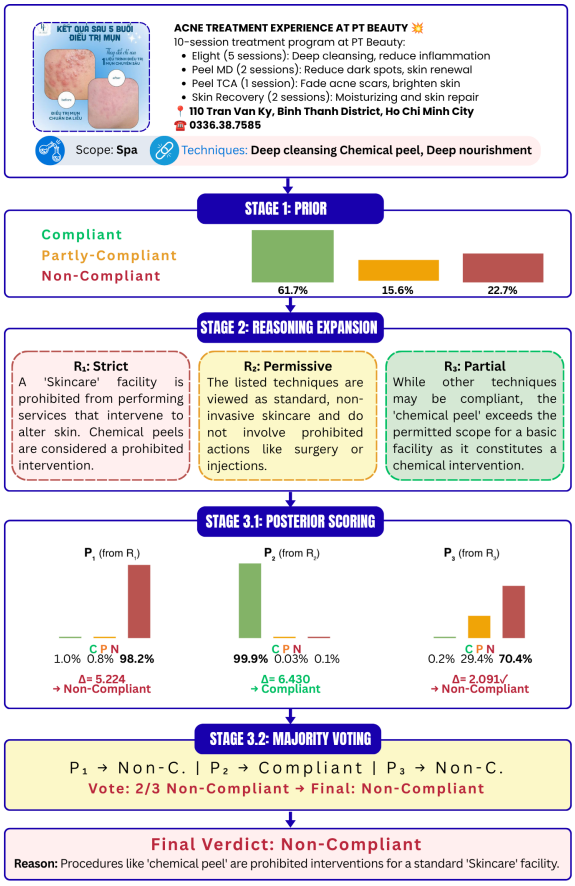


Figure 3: Step-by-step causal evaluation of an evasive advertisement showing the contradiction between the "non-invasive" text and the injection image.

trajectories. Crucially, the Evidence Adjudicator evaluates the log-odds delta (Δ) triggered by this retrieved evidence for each path. By isolating the explicit legal violation against the previously assumed benign context, the posterior probability for non-compliance surges to 98.2%. The resulting Δ significantly eclipses the strict liability threshold ($\tau = 0.5$), enabling the framework to systematically prune hallucinated compliant branches (such as the erroneously permissive R_2) and demonstrating profound resilience against contextually disguised regulatory breaches.

5.5 Analysis and Discussion

5.5.1 Benchmarking Explanation Quality. In regulatory contexts, the justification $\boldsymbol{X}$ is as critical as the verdict $\boldsymbol{y}$. Evaluating the generated explanations using the multi-dimensional FIDES-Score (see Table 5), MM-RegCheck achieves the highest overall score of 3.71, outperforming Ministral (3.56), Qwen3-VL (3.35), and Llama-4 (3.00). The framework excels particularly in the *Justification* dimension, as the explanations generated by the surviving TF-BEAR branches strictly cite the retrieved regulatory constraints ($\hat{C}$) rather than relying on ungrounded generative assumptions.

Table 5: Evaluation of generated explanations using the multi-dimensional FIDES-Score: Fidelity (F), Justification (J), and Clarity (C). Bold indicates the best performance, while underline denotes the second-best.

Model	F	J	C	FIDES
Phi-4 (14B)	2.4	2.7	3.5	2.9
Ministral (14B)	2.9	3.5	4.3	3.6
Llama-4-Maverick (17B)	2.4	2.7	3.9	3.0
Qwen3-VL (8B)	2.8	3.2	4.0	3.4
Qwen3.5 9B (Ours)	**3.1**	**3.7**	**4.4**	**3.7**

5.5.2 Cross-Modality Generalization. To demonstrate the generalizability of TF-BEAR, we performed a backward compatibility test, adapting the framework to the *VietCheckMed* dataset [29]. Our proposed engine achieved an F1-score of 0.8920 in Beauty and 0.6794 in Healthcare. Remarkably, this surpasses the previous SOTA established by the massive Llama-3.3 (70B) model, which scored 0.8698 and 0.6624, respectively (Figure 4). This confirms that our methodology transcends modality limitations, extracting superior reasoning efficiency from significantly smaller parameter models.

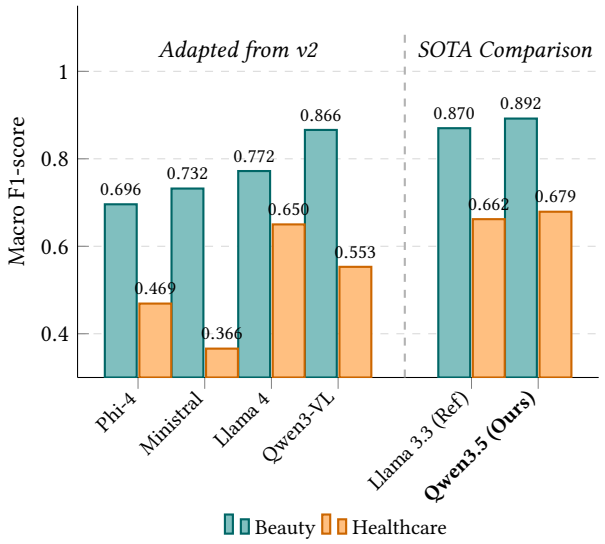


Figure 4: Backward compatibility performance on the *VietCheckMed* dataset. A clear separation is made between models adapted from the multimodal setting and the established baseline (Llama 3.3). Our MM-RegCheck (Qwen3.5 9B) effectively surpasses the reference SOTA.

5.5.3 Qualitative Error Analysis. To identify the boundaries of our framework, we conducted an audit of residual misclassifications, categorizing them into four primary modes (Table 6). **First, Visual-Semantic Mismatch (43%)** constitutes the primary bottleneck, predominantly occurring in the Healthcare domain. In these instances, the VLM fails to semantically align highly nuanced medical slang with rigid legal terminology. **Second, Inference Hallucination (36%)** occurs when the model correctly identifies the procedure and retrieves the statute, but incorrectly deduces compliance. **Third, Context Retrieval Failure (18%)** arises when the dense vector search inadvertently retrieves legally proximate but functionally incorrect operational scopes. **Finally, Technique Extraction Error (3%)** accounts for a remarkably low percentage of failures, empirically validating the robustness of our concept-guided extraction prompt in isolating physical acts over marketing claims.

Table 6: Distribution of Error Types.

Error Category	Percentage (%)
Visual-Semantic Mismatch	43%
Inference Hallucination	36%
Context Retrieval Failure	18%
Technique Extraction Error	3%

6 CONCLUSION & FUTURE WORK

This paper introduces MM-RegCheck, establishing a new paradigm for automated oversight in digital advertising by shifting the core objective from traditional veracity checking to explicit, authorization-based regulatory compliance. By elegantly integrating Retrieval-Augmented Generation with the Training-Free Bayesian Evidence Aggregation for Reliable LLM Reasoning (TF-BEAR) algorithm, we successfully mitigate multimodal hallucinations and adversarial cross-modal evasion tactics. Extensive evaluations on the expertly validated MedReg-MM benchmark demonstrate that our highly efficient 9B-parameter framework establishes a definitive new SOTA. It achieves a peak Macro-F1 score of 0.8985 under standard conditions and robustly outperforms massive architectures (e.g., Llama-4) in complex adversarial environments, all while operating at an exceptionally scalable computational cost. Ultimately, MM-RegCheck provides a mathematically rigorous, highly efficient, and zero-day adaptable solution for high-stakes regulatory monitoring.

Guided by the quantitative insights from our component-level failure audit (detailed in Section 5.5), future work must evolve along three highly targeted scientific trajectories to address residual bottlenecks. To resolve the pervasive visual-semantic mismatch, future iterations should transition from standard dense vector retrieval to Legal Knowledge Graph-augmented RAG (KG-RAG). Integrating topological reasoning over structured legal ontologies has been shown to significantly enhance multi-hop semantic alignment in specialized domains [24]. Furthermore, to combat inference hallucinations triggered by complex legal phrasing, the evidence adjudicator can be expanded utilizing Multi-Agent Debate frameworks. By instantiating multiple LLM agents with opposing legal perspectives to iteratively critique the retrieved evidence, the framework can structurally resolve double-negative logic failures prior to final aggregation [8, 19]. Finally, adapting this static framework to the rapid proliferation of short-form video content requires incorporating spatiotemporal causal reasoning. This involves modeling temporal confounders to detect fleeting regulatory violations intentionally obscured within dynamic video sequences [21].

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References

[1] Marah Abdin, Jyoti Aneja, Harkirat Behl, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, Michael Harrison, Russell J Hewett, Mojan Javaheripi, Piero Kauffmann, et al. 2024. Phi-4 technical report. *arXiv preprint arXiv:2412.08905* (2024).

[2] Esma Aïmeur, Sabrine Amri, and Gilles Brassard. 2023. Fake news, disinformation and misinformation in social media: a review. *Social Network Analysis and Mining* 13, 1 (2023), 30.

[3] Isabelle Augenstein, Christina Lioma, Dongsheng Wang, Lucas Chaves Lima, Casper Hansen, Christian Hansen, and Jakob Grue Simonsen. 2019. MultiFC: A Real-World Multi-Domain Dataset for Evidence-Based Fact Checking of Claims. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (Eds.). Association for Computational Linguistics, Hong Kong, China, 4685–4697. doi:10.18653/v1/D19-1475

[4] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609* (2023).

[5] Canyu Chen, Haoran Wang, Matthew Shapiro, Yunyu Xiao, Fei Wang, and Kai Shu. 2022. Combating health misinformation in social media: Characterization, detection, intervention, and open issues. *arXiv preprint arXiv:2211.05289* (2022).

[6] Wenliang Dai, Zihan Liu, Ziwei Ji, Dan Su, and Pascale Fung. 2023. Plausible May Not Be Faithful: Probing Object Hallucination in Vision-Language Pre-training. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, Andreas Vlachos and Isabelle Augenstein (Eds.). Association for Computational Linguistics, Dubrovnik, Croatia, 2136–2148. doi:10.18653/v1/2023.eacl-main.156

[7] Marina Danilevsky, Kun Qian, Ranit Aharonov, Yannis Katsis, Ban Kawas, and Prithviraj Sen. 2020. A Survey of the State of Explainable AI for Natural Language Processing. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, Kam-Fai Wong, Kevin Knight, and Hua Wu (Eds.). Association for Computational Linguistics, Suzhou, China, 447–459. doi:10.18653/v1/2020.aacl-main.46

[8] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*. JMLR.org, Vienna, Austria, Article 467, 31 pages.

[9] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences* 120, 30 (2023), e2305016120.

[10] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6904–6913.

[11] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).

[12] Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. A survey on automated fact-checking. *Transactions of the association for computational linguistics* 10 (2022), 178–206.

[13] Alon Jacovi and Yoav Goldberg. 2020. Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness?. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 4198–4205. doi:10.18653/v1/2020.acl-main.386

[14] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM computing surveys* 55, 12 (2023), 1–38.

[15] Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. 2024. Mixtral of experts. *arXiv preprint arXiv:2401.04088* (2024).

[16] Payam Karisani and Heng Ji. 2024. Fact checking beyond training set. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 2247–2261.

[17] Divyansh Kaushik, Amrith Setlur, Eduard Hovy, and Zachary C Lipton. 2021. Explaining the Efficacy of Counterfactually Augmented Data. *International Conference on Learning Representations (ICLR)* (2021).

[18] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.

[19] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 conference on empirical methods in natural language processing*. 17889–17904.

[20] Xubo Liu, Wenya Guo, Ruxue Yan, Xumeng Liu, Ying Zhang, and Ru Zhou. 2025. Rethinking the Reliability of Evidence in End-to-End Fact-Checking from the Causal Perspective. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 9044–9052.

[21] Jiahao Meng, Xiangtai Li, Haochen Wang, Yue Tan, Tao Zhang, Lingdong Kong, Yunhai Tong, Anran Wang, Zhiyang Teng, Yujing Wang, et al. 2025. Open-o3 Video: Grounded Video Reasoning with Explicit Spatio-Temporal Evidence. *arXiv preprint arXiv:2510.20579* (2025).

[22] Kai Nakamura, Sharon Levy, and William Yang Wang. 2020. Fakeddit: A new multimodal benchmark dataset for fine-grained fake news detection. In *Proceedings of the twelfth language resources and evaluation conference*. 6149–6157.

[23] Yulei Niu, Kaihua Tang, Hanwang Zhang, Zhiwu Lu, Xian-Sheng Hua, and Ji-Rong Wen. 2021. Counterfactual vqa: A cause-effect look at language bias. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12700–12710.

[24] Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Jiapu Wang, and Xindong Wu. 2024. Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering* 36, 7 (2024), 3580–3599.

[25] Judea Pearl. 2009. Causal inference in statistics: An overview. (2009).

[26] Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases?. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*. 2463–2473.

[27] S Suryavardan, Shreyash Mishra, Parth Patwa, Megha Chakraborty, Anku Rani, Aishwarya Reganti, Aman Chadha, Amitava Das, Amit Sheth, Manoj Chinnakotla, et al. 2023. Factify 2: A multimodal fake news and satire news dataset. *arXiv preprint arXiv:2304.03897* (2023).

[28] Sahar Tahmasebi, Eric Müller-Budack, and Ralph Ewerth. 2024. Multimodal misinformation detection using large vision-language models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 2189–2199.

[29] Nguyen Thanh Tam, Khanh Quoc Tran, Dat Thanh Pham, Truong Phu Le, Nguyen Hoang Gia Han, and Binh T Nguyen. 2026. VietCheckMed: Explainable Regulatory Compliance Checking for Medical Advertisements on Vietnamese Social Media. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 1069–1077.

[30] Andreas Vlachos and Sebastian Riedel. 2014. Fact checking: Task definition and dataset construction. In *Proceedings of the ACL 2014 workshop on language technologies and computational social science*. 18–22.

[31] An Vo, Khai-Nguyen Nguyen, Mohammad Reza Taesiri, Vy Tuong Dang, Anh Totti Nguyen, and Daeyoung Kim. 2025. Vision language models are biased. *arXiv preprint arXiv:2505.23941* (2025).

[32] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.

[33] Wenyan Xu, Dawei Xiang, Tianqi Ding, and Weihai Lu. 2025. MMM-Fact: A Multimodal, Multi-Domain Fact-Checking Dataset with Multi-Level Retrieval Difficulty. arXiv:2510.25120 [cs.SI] https://arxiv.org/abs/2510.25120

[34] Barry Menglong Yao, Aditya Shah, Lichao Sun, Jin-Hee Cho, and Lifu Huang. 2023. End-to-end multimodal fact-checking and explanation generation: A challenging dataset and models. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Taipei, Taiwan, 2733–2743.

[35] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems* 36 (2023), 11809–11822.

[36] Dimitrina Zlatkova, Preslav Nakov, and Ivan Koychev. 2019. Fact-checking meets fauxtography: Verifying claims about images. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 2099–2108.